COURSE OUTLINE

Department:	Faculty of Computing	**Knowledge Group:**		Artificial Intelligence and Machine Learning	
Programme:	Software Engineering	**Class:**	BESE-13A		
Course code:	CS-471	**Academic Session/Semester:**		Spring 2025	
Course name:	Machine Learning	**Pre/co requisite (course name and code, if applicable):**		None	
Credit hours:	3+1				

Course Synopsis	This course is designed to provide students with a comprehensive and in-depth understanding of the core principles, techniques, and applications of machine learning. This course explores the foundations of supervised, unsupervised, and reinforcement learning, delving into topics such as classification, regression, clustering, and neural networks. The knowledge gained in this course will enable the students to build and evaluate predictive models, extract meaningful insights from data, and make informed decisions in various domains, from business analytics to healthcare. Through a combination of theoretical knowledge and practical application, this course equips students with the skills and expertise needed to excel in the dynamic and rapidly evolving field of machine learning.			
Course Learning Outcomes (CLOs)	At the end of the course, the students will be able to: 1. **Apply** various machine learning algorithms, tools and techniques to **solve** a given problem. 2. **Analyse** a given problem and select the most suitable machine learning technique for the task. 3. **Develop** solutions for practical challenges by employing machine learning algorithms. 4. **Implement** various machine learning algorithms using modern software libraries for a range of applications.			
Course Schedule	Thursday 2 to 4 pm (CR-12) Friday 12 to 1 pm (CR-12)			
Course lecturer	**Name**	**Office**	**Contact no.**	**E-mail**
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Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	Course Learning Outcomes	PLO (SE)	PLO (CS)	BT Level	Teaching & Learning Methods	Assessment Methods
CLO 1	**Apply** various machine learning algorithms, tools and techniques to **solve** a given problem.	PLO-1	NA	C-3	Active Learning, Blended	Quiz Assignment MSE

					Learning	ESE
CLO 2	**Analyse** a given problem and select the most suitable machine learning technique for the task.	PLO-2	NA	C-4	Active Learning, Blended Learning	Quiz Assignment MSE ESE
CLO 3	**Develop** solutions for practical challenges by employing machine learning algorithms.	PLO-3	NA	C-6	Active Learning, Cooperative Learning, Blended Learning	Quiz Assignment MSE ESE Lab Work Project
CLO 4	**Implement** various machine learning algorithms using modern software libraries for a range of applications.	PLO-5	NA	P-4	Active Learning, Cooperative Learning, Blended Learning	Lab Work Project

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class or lab activities.
2.	Cooperative learning	Conducted through design project. Students in a team of two to three will be given a design project that requires software engineering process design solutions. Compliance to the design specifications need to be given in the form of written reports.
3.	Blended learning	Conducted through Learning Management System (LMS) of NUST. All information as well as materials related to teaching and learning activities will be shared with class through this system. Some for formative assessments will be also conducted using this system.

Weekly Schedule (Tentative):

Week	Topics	Assessment Plan
Week 1	Introduction to the course, ML Landscape	
Week 2	Problem Formulation & Evaluation, Data Splits and Validation	
Week 3	Linear Regression, Gradient Descent	Quiz 1
Week 4	Logistic Regression	Assignment 1
Week 5	Decision Trees and Ensemble Models (Random Forest)	Quiz 2

Week 6	KNN and Parzen Window	Assignment 2, Project Floated
Week 7	Support Vector Machines and kernel tricks	Quiz 3
Week 8	Debugging ML and Error Analysis	Project Proposal Submission
Week 9	**Mid-Semester Exam**	
Week 10	Generative Reasoning Algorithms (Naive Bayes, Gaussian Discriminant Analysis)	
Week 11	Neural Networks, MLPs and Generalised Linear Models.	
Week 12	Clustering Algorithms	Assignment 3, Quiz 4
Week 13	PCA, Dimensionality Reduction	
Week 14	Reinforcement Learning	Quiz 5
Week 15	ML Models Deployment and MLOps	
Week 16	Advanced Topics in Machine Learning.	Quiz 6, Project Report Submission
Week 17	Project Presentations	
Week 18	**End Semester Exam**	

Lab Experiments (if applicable):

Lab 1	Introduction to scikit learn, pandas, and numpy
Lab 2	Model selection and experiment design
Lab 3	Linear Regression
Lab 4	Logistic Regression
Lab 5	Decision Trees
Lab 6	K Nearest Neighbours
Lab 7	Support Vector Machines
Lab 8	Open Ended Lab
Week 9	**Mid-Semester Break**
Lab 9	Naïve Bayes Classifier
Lab 10	Artificial Neural Networks and Backpropagation
Lab 11	Clustering Techniques
Lab 12	Dimensionality Reduction
Lab 13	Reinforcement Learning
Lab 14	Introduction to MLOps
Lab 15	Open Ended Lab
Lab 16	Project Demonstration and Presentations
Week 18	**End Semester Break**

Assessment Methods:

Assessment		Percentage
Theory: 75%		
1	Quizzes (10-15%)	10%
2	Assignments (5-10%)	5%
3	Mid-Term Exam (25-35%)	30%
4	End-Semester Exam (40-50%)	45%
5	Project	10%
Labs:25%		
1	Labs	60%
2	Project	20%
3	Final Lab	20%
Total:		**100%**

Learning resources:

Textbook:

1. RS Michalski, JG Carbonell, TM Mitchell, "Machine Learning: An Artificial Intelligence Approach", Springer, 2014
2. Christopher M. Bishop, "Pattern Recognition and Machine Learning", 2007

Reference Book:

1. Richard Duda, Peter Hart and David Stork, Pattern Classification, 2nd ed. John Wiley & Sons, 2001.
2. Richard Sutton and Andrew Barto, Reinforcement Learning: An introduction. MIT Press, 1998

Grading Policy:

Quiz Policy:

The quizzes will be unannounced / announced and normally last for ten minutes. The question framed is to test the concepts involved in last few lectures. Number of quizzes that will be used for evaluation is at the instructor's discretion.

Project Policy:

Students will be required to develop a project during the course which should be completed towards the end of the semester. They will be graded based on project deliverables and presentation at the end. Students will work in a group/team for projects. A group of 3 students is recommended.

Assignment Policy:

In order to develop comprehensive understanding of the subject, assignments will be given. Late assignments will not be accepted / graded. All assignments will count towards the total (No 'best-of' policy). The students are advised to do the assignment themselves. Copying of assignments is highly discouraged and violations will be dealt with severely by referring any occurrences to the disciplinary committee.

Plagiarism:

SEECS maintains a zero-tolerance policy towards plagiarism. While collaboration in this course is highly encouraged, you must ensure that you do not claim other people's work/ ideas as your own. Plagiarism occurs when the words, ideas, assertions, theories, figures, images, programming codes of others are presented as your own work. You must cite and acknowledge all sources of information in your assignments. Failing to comply with the SEECS plagiarism policy will lead to strict penalties including zero marks in assignments and referral to the academic coordination office for disciplinary action.